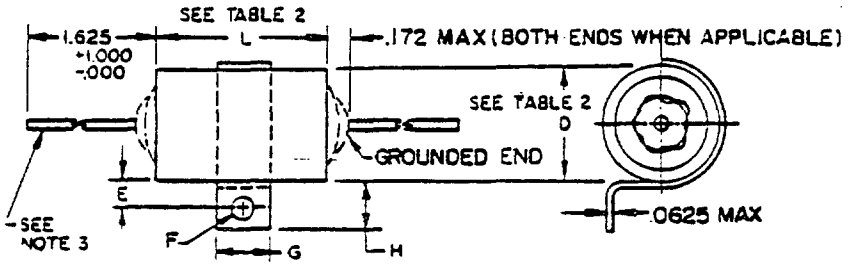
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, PLASTIC (OR PAPER-PLASTIC) DIELECTRIC, AXIAL-WIRE TERMINAL, TANGENTIAL RETAINER, TUBULAR (UNINSULATED) (HERMETICALLY SEALED IN METAL CASES) ESTABLISHED RELIABILITY, STYLE CQR42

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-19978.

INACTIVE FOR NEW EQUIPMENT DESIGN AFTER DATE OF THIS SPECIFICATION SHEET



INCHES	MM	INCHES	MM	INCHES	MM
.001	.03	.0625	1.59	.312	7.92
.004	.10	.144	3.66	.400	10.16
.005	.13	.156	3.96	.438	11.13
.025	.64	.172	4.37	.500	12.70
.031	.79	.188	4.78	.562	14.27
.032	.81	.235	5.97	1.000	25.40
.062	1.57	.250	6.35	1.625	41.28

Case dimension D	H = .062	G = .062	E = .031	F = .005
Less than .562	.312	.250	.188	.144
.562 and up	.406	.500	.250	.156

NOTES:

1. **Dimensions are in inches.**
2. **Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.**
3. **Leads shall be of tinned solid wire, .025(No. 22 AWG) for cases .235 and .312 in diameter; and .032(No. 20 AWG) for cases .400 diameter and above. Tolerance on all lead wire diameters shall be +.004, -.001.**

FIGURE 1. Style CQR42 capacitors.

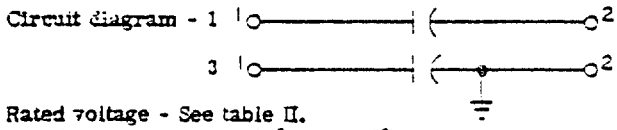
FSC 5910

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REQUIREMENTS:

Design and construction:
Dimensions and configuration - See figure 1 and table II.
Case material - Nonmagnetic (end seal may be of magnetic material).
Dielectric material - Paper and polyethelene terephthalate (K).
Terminals - Axial-wire lead (see figure 1).

Circuit diagram - 1



Rated voltage - See table II.
Rated temperature - -65° to +125° C.
Capacitance (Cap.) (nom) - See table II.
Capacitance tolerance - K, ±10 percent.
Dissipation factor (DF) (max) - 1.0 percent.
Failure rate level - In accordance with MIL-C-19978.
Reliability - In accordance with MIL-C-19978.

Burn-in: In accordance with MIL-C-19978.

Radiographic inspection: In accordance with MIL-C-19978.

Seal: Method 112 of MIL-STD-202, test condition letter - A.

Barometric pressure: In accordance with MIL-C-19978.
Test points:
Circuit diagram 1 - Between terminals and case.
Circuit diagram 3 - Between ungrounded terminal and case.

Insulation resistance (IR):
Terminal to terminal - See table I.

TABLE I. Terminal-to-terminal insulation resistance.

Capacitance rating	Minimum insulation resistance
Characteristic K	
	At 25° C
0 to 0.6 microfarad	25,000 megohms
Greater than 0.6 microfarad	15,000 megohm-microfarads 1/
	At 125° C
0 to 0.08 microfarad	250 megohms
Greater than 0.08 microfarad	20 megohm-microfarads 1/

1/ Product obtained by multiplying the capacitance in microfarads by the insulation resistance in megonms.

Terminal to case - Greater than 10,000 megohms.

Vibration, high frequency: Method 204 of MIL-STD-202, test condition B, with the following exception:
Direction and duration of motion - 4 hours in each of two mutually perpendicular directions (total of 8 hours), one parallel and the other perpendicular to the cylindrical axis.

Salt spray: In accordance with MIL-C-19978.

Solderability: In accordance with MIL-C-19978.

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TABLE II. Style CQR42 capacitors.

Part number 1/	DC voltage rating	Capacitance rating	Failure rate level	Case dimensions			
				Characteristic K			
				Circuit 1		Circuit 3	
				L±.031	D +.015 -.005	L±.031	D +.015 -.005
	Volts	µF		Inches	Inches	Inches	Inches
CQR42A-KB822K3-	100	.0082	L, M, P	.750	.235	.688	.235
CQR42A-KB103K3-	100	.010		.750	.235	.688	.235
CQR42A-KB123K3-	100	.012		.750	.235	.688	.235
CQR42A-KB153K3-	100	.015		.750	.235	.688	.235
CQR42A-KB393K3-	100	.039		.875	.312	.812	.312
CQR42A-KB473K3-	100	.047		.875	.312	.812	.312
CQR42A-KB823K3-	100	.082		.875	.400	.812	.400
CQR42A-KB104K3-	100	.10		.875	.400	.812	.400
CQR42A-KB124K3-	100	.12		1.125	.400	1.062	.400
CQR42A-KB154K3-	100	.15		1.125	.400	1.062	.400
CQR42A-KB184K3-	100	.18		1.375	.400	1.312	.400
CQR42A-KB224K3-	100	.22		1.375	.400	1.312	.400
CQR42A-KB274K3-	100	.27		1.125	.562	1.062	.562
CQR42A-KB334K3-	100	.33		1.125	.562	1.062	.562
CQR42A-KB394K3-	100	.39		1.375	.562	1.312	.562
CQR42A-KB474K3-	100	.47		1.375	.562	1.312	.562
CQR42A-KB564K3-	100	.56		1.625	.562	1.562	.562
CQR42A-KB684K3-	100	.68		1.625	.562	1.562	.562
CQR42A-KB824K3-	100	.82		1.625	.670	1.562	.670
CQR42A-KB105K3-	100	1.0		1.625	.670	1.562	.670
CQR42A-KC392K3-	200	.0039		.750	.235	.688	.235
CQR42A-KC472K3-	200	.0047		.750	.235	.688	.235
CQR42A-KC562K3-	200	.0056		.750	.235	.688	.235
CQR42A-KC682K3-	200	.0068		.750	.235	.688	.235
CQR42A-KC183K3-	200	.018		.875	.312	.812	.312
CQR42A-KC223K3-	200	.022		.875	.312	.812	.312
CQR42A-KC273K3-	200	.027		.875	.312	.812	.312
CQR42A-KC333K3-	200	.033		.875	.312	.812	.312
CQR42A-KC393K3-	200	.039		.875	.400	.812	.400
CQR42A-KC473K3-	200	.047		.875	.400	.812	.400
CQR42A-KC563K3-	200	.056		.875	.400	.812	.400
CQR42A-KC683K3-	200	.068		.875	.400	.812	.400
CQR42A-KC823K3-	200	.082		1.125	.400	1.062	.400
CQR42A-KC104K3-	200	.10		1.125	.400	1.062	.400
CQR42A-KC124K3-	200	.12		1.375	.400	1.312	.400
CQR42A-KC154K3-	200	.15		1.375	.400	1.312	.400
CQR42A-KC184K3-	200	.18		1.125	.562	1.062	.562
CQR42A-KC224K3-	200	.22		1.125	.562	1.062	.562
CQR42A-KC274K3-	200	.27		1.375	.562	1.312	.562
CQR42A-KC334K3-	200	.33		1.375	.562	1.312	.562
CQR42A-KC394K3-	200	.39		1.625	.562	1.562	.562
CQR42A-KC474K3-	200	.47		1.625	.562	1.562	.562
CQR42A-KC564K3-	200	.56		1.625	.670	1.562	.670
CQR42A-KC684K3-	200	.68		1.625	.670	1.562	.670
CQR42A-KC824K3-	200	.82		2.125	.750	2.062	.750
CQR42A-KC105K3-	200	1.0	L, M, P	2.125	.750	2.062	.750

See footnote at end of table.

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TABLE II. Style CQR42 capacitors. -Continued

Part number 1/	DC voltage rating	Capacitance rating	Failure rate level	Case dimensions			
				Characteristic K			
				Circuit 1		Circuit 3	
				L ± .031	D +.015 -.005	L ± .031	D +.015 -.005
	Volts	µF		Inches	Inches	Inches	Inches
CQR42A-KE272K3-	400	.0027	L. M. P	.750	.235	.688	.235
CQR42A-KE332K3-	400	.0033		.750	.235	.688	.235
CQR42A-KE123K3-	400	.012		.875	.312	.812	.312
CQR42A-KE153K3-	400	.015		.875	.312	.812	.312
CQR42A-KE273K3-	400	.027		.875	.400	.812	.400
CQR42A-KE333K3-	400	.033		.875	.400	.812	.400
CQR42A-KE393K3-	400	.039		1.125	.400	1.062	.400
CQR42A-KE473K3-	400	.047		1.125	.400	1.062	.400
CQR42A-KE563K3-	400	.056		1.375	.400	1.312	.400
CQR42A-KE683K3-	400	.068		1.375	.400	1.312	.400
CQR42A-KE823K3-	400	.082		1.125	.562	1.062	.562
CQR42A-KE104K3-	400	.10		1.125	.562	1.062	.562
CQR42A-KE124K3-	400	.12		1.375	.562	1.312	.562
CQR42A-KE154K3-	400	.15		1.375	.562	1.312	.562
CQR42A-KE184K3-	400	.18		1.625	.562	1.562	.562
CQR42A-KE224K3-	400	.22		1.625	.562	1.562	.562
CQR42A-KE274K3-	400	.27		1.625	.670	1.562	.670
CQR42A-KE334K3-	400	.33		1.625	.670	1.562	.670
CQR42A-KE394K3-	400	.39		2.125	.750	2.062	.750
CQR42A-KE474K3-	400	.47		2.125	.750	2.062	.750
CQR42A-KF102K3-	600	.0010		.750	.235	.688	.235
CQR42A-KF122K3-	600	.0012		.750	.235	.688	.235
CQR42A-KF152K3-	600	.0015		.750	.235	.688	.235
CQR42A-KF182K3-	600	.0018		.750	.235	.688	.235
CQR42A-KF222K3-	600	.0022		.750	.235	.688	.235
CQR42A-KF272K3-	600	.0027		.875	.312	.312	.312
CQR42A-KF332K3-	600	.0033		.875	.312	.812	.312
CQR42A-KF392K3-	600	.0039		.875	.312	.812	.312
CQR42A-KF472K3-	600	.0047		.875	.312	.812	.312
CQR42A-KF562K3-	600	.0056		.875	.312	.812	.312
CQR42A-KF682K3-	600	.0068		.875	.312	.812	.312
CQR42A-KF822K3-	600	.0082		.875	.312	.812	.312
CQR42A-KF103K3-	600	.010		.875	.312	.812	.312
CQR42A-KF123K3-	600	.012		.875	.400	.812	.400
CQR42A-KF153K3-	600	.015		.875	.400	.812	.400
CQR42A-KF183K3-	600	.018		.875	.400	.812	.400
CQR42A-KF223K3-	600	.022		.875	.400	.812	.400
CQR42A-KF273K3-	600	.027		1.125	.400	1.062	.400
CQR42A-KF333K3-	600	.033		1.125	.400	1.062	.400
CQR42A-KF393K3-	600	.039		1.375	.400	1.312	.400
CQR42A-KF473K3-	600	.047		1.375	.400	1.312	.400
CQR42A-KF563K3-	600	.056		1.125	.562	1.062	.562
CQR42A-KF683K3-	600	.068		1.125	.562	1.062	.562
CQR42A-KF823K3-	600	.082		1.375	.562	1.312	.562
CQR42A-KF104K3-	600	.10		1.375	.562	1.312	.562
CQR42A-KF124K3-	600	.12	L. M. P	1.625	.562	1.562	.562

See footnote at end of table.

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TABLE II. Style CQR42 capacitors. -Continued

Part number 1/	DC voltage rating	Capacitance rating	Failure rate level	Case dimensions			
				Characteristic K			
				Circuit 1		Circuit 3	
				L±.031	D +.015 -.005	L±.031	D +.015 -.005
	Volts	µF		Inches	Inches	Inches	Inches
CQR42A-KF154K3-	600	.15	L, M, P	1.625	.562	1.562	.562
CQR42A-KF184K3-	600	.18		1.625	.670	1.562	.670
CQR42A-KF224K3-	600	.22		1.625	.670	1.562	.670
CQR42A-KF274K3-	600	.27		2.125	.750	2.062	.750
CQR42A-KF334K3-	600	.33		2.125	.750	2.062	.750
CQR42A-KF394K3-	600	.39		2.375	.750	2.312	.750
CQR42A-KF474K3-	600	.47		2.375	.750	2.312	.750
CQR42A-KG102K3-	1,000	.001		.875	.400	.812	.400
CQR42A-KG122K3-	1,000	.0012		.875	.400	.812	.400
CQR42A-KG152K3-	1,000	.0015		.875	.400	.812	.400
CQR42A-KG182K3-	1,000	.0018		.875	.400	.812	.400
CQR42A-KG222K3-	1,000	.0022		.875	.400	.812	.400
CQR42A-KG272K3-	1,000	.0027		.875	.400	.812	.400
CQR42A-KG332K3-	1,000	.0033		.875	.400	.812	.400
CQR42A-KG392K3-	1,000	.0039		.875	.400	.812	.400
CQR42A-KG472K3-	1,000	.0047		.875	.400	.812	.400
CQR42A-KG562K3-	1,000	.0056		.875	.400	.812	.400
CQR42A-KG682K3-	1,000	.0068		.875	.400	.812	.400
CQR42A-KG822K3-	1,000	.0082		.875	.400	.812	.400
CQR42A-KG103K3-	1,000	.01		.875	.400	.812	.400
CQR42A-KG123K3-	1,000	.012		1.125	.400	1.062	.400
CQR42A-KG153K3-	1,000	.015		1.125	.400	1.062	.400
CQR42A-KG183K3-	1,000	.018		1.375	.400	1.312	.400
CQR42A-KG223K3-	1,000	.022		1.375	.400	1.312	.400
CQR42A-KG273K3-	1,000	.027		1.125	.562	1.062	.562
CQR42A-KG333K3-	1,000	.033		1.125	.562	1.062	.562
CQR42A-KG393K3-	1,000	.039		1.375	.562	1.312	.562
CQR42A-KG473K3-	1,000	.047		1.375	.562	1.312	.562
CQR42A-KG563K3-	1,000	.056		1.625	.562	1.562	.562
CQR42A-KG683K3-	1,000	.068		1.625	.562	1.562	.562
CQR42A-KG823K3-	1,000	.082		1.625	.670	1.562	.670
CQR42A-KG104K3-	1,000	.10		1.625	.670	1.562	.670
CQR42A-KG124K3-	1,000	.12		1.875	.670	1.812	.670
CQR42A-KG154K3-	1,000	.15		1.875	.670	1.812	.670
CQR42A-KG184K3-	1,000	.18		2.125	.750	2.062	.750
CQR42A-KG224K3-	1,000	.22		2.125	.750	2.062	.750
CQR42A-KG274K3-	1,000	.27		2.125	1.000	2.062	1.000
CQR42A-KG334K3-	1,000	.33		2.125	1.000	2.062	1.000
CQR42A-KG394K3-	1,000	.39		2.375	1.000	2.312	1.000
CQR42A-KG474K3-	1,000	.47	L, M, P	2.375	1.000	2.312	1.000

1/ Complete part number shall include additional symbols to indicate circuit and failure rate level, as applicable.

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TABLE III. Millimeter equivalents to decimal inches.

INCHES	MM	INCHES	MM
0.005	0.13	1.000	25.40
0.015	0.38	1.062	26.97
0.031	0.79	1.125	28.58
0.235	5.97	1.312	33.32
0.312	7.92	1.375	34.93
0.400	10.16	1.562	39.67
0.562	14.27	1.625	41.28
0.670	17.02	1.812	46.02
0.688	17.48	2.062	52.37
0.750	19.05	2.125	53.98
0.812	20.62	2.312	58.72
0.875	22.23	2.375	60.33

Terminal strength: Method 211 of MIL-STD-202, test condition letter - D.

Stability at low and high temperature:
Low temperature:
Test temperature - $-65^{\circ}\ ^{+0^{\circ}}_{-5^{\circ}}$ C.

Capacitance change (max) - -10 percent.
High temperature:
Test temperature - $+125^{\circ}\ ^{+5^{\circ}}_{-0^{\circ}}$ C.

Capacitance change (max) - +10 percent.

Life:
Capacitance change (max) - ±5 percent.

Resistance to soldering heat: In accordance with MIL-C-19978.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army - ML, MU
Navy - AS
Air Force - 11, 17
DSA - ES

User activities:
Army - ME, AT, AV
Navy - MC, AS, OS, SH
Air Force -

Preparing activity:
Air Force - 80

Agent:
DSA - ES

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